

Java Full Stack Interview Roadmap (Core Java First)

What to Learn First

Core Java (Modules 1–9), then Advanced Java (JDBC, Servlets, JSP), then Spring Core, Spring Boot, Hibernate/JPA, REST APIs, MySQL, Maven/Gradle, Git/GitHub, Microservices (optional), Docker & AWS (optional).

Module 1: Java Fundamentals

Java, History, Features, Java Editions, Java Versions, JDK, JRE, JVM, Java Architecture, Program Execution, Bytecode, Interpreter, JIT, AOT, JVM Memory, Heap, Stack, Metaspace, PC Register, Class Loader, StackOverflowError, OutOfMemoryError, Keywords, Identifiers, Naming Conventions

Module 2: Variables & Language Basics

Variables, Local/Instance/Static Variables, Primitive & Reference Types, Default Values, Literals, Unicode, ASCII, Escape Sequences, Number Systems, Operators, Operator Precedence, Type Casting, Type Promotion, Scanner, BufferedReader, Command Line Arguments, Math Class, System Class, Control Statements, Loops, break, continue, return

Module 3: Methods & Memory

Methods, Method Signature, Overloading, Recursion, Pass by Value, Stack Frames, Static Methods, Static Block, Static Variables, Constructors, Constructor Chaining, this, super, Packages, Import, Static Import, varargs, enum, Access Modifiers

Module 4: OOP

Class, Object, Encapsulation, Inheritance, Polymorphism, Overriding, Dynamic & Static Binding, Abstract Class, Interface, Default/Static/Private Interface Methods, Association, Aggregation, Composition, IS-A, HAS-A, final, Object Class, equals(), hashCode(), toString(), clone(), Immutable Class, Inner Classes, SOLID (Basic)

Module 5: Arrays & Strings

Arrays, Multi-dimensional Arrays, Jagged Arrays, Arrays Utility, String, SCP, StringBuilder, StringBuffer, intern(), Wrapper Classes, Autoboxing, Unboxing, Integer Cache, StringTokenizer, StringJoiner, Deep Copy, Shallow Copy, Regex, printf()

Module 6: Exception Handling

Throwable, Exception, Error, Checked/Unchecked Exceptions, try, catch, finally, throw, throws, Multi-catch, Nested try, Exception Propagation, Exception Chaining, printStackTrace(), try-with-resources, AutoCloseable, Custom Exceptions

Module 7: Collections

Collection Framework, Hierarchy, Iterable, Iterator, ListIterator, Enumeration, List, ArrayList, LinkedList, Vector, Stack, Queue, PriorityQueue, Deque, ArrayDeque, Set, HashSet, LinkedHashSet, TreeSet, Map, HashMap, LinkedHashMap, TreeMap, WeakHashMap, IdentityHashMap, Comparable, Comparator, Collections Class, Arrays Class, HashMap Internals, Time Complexity

Module 8: Multithreading

Process vs Thread, Lifecycle, Thread, Runnable, Thread Methods, Synchronization, synchronized, wait(), notify(), notifyAll(), Deadlock, volatile, Atomic Classes, Executor Framework, Thread Pool, Callable, Future, CompletableFuture, ConcurrentHashMap, Thread Priority, Daemon Thread

Module 9: File Handling & Serialization

File, FileReader, FileWriter, BufferedReader, BufferedWriter, Serialization, Deserialization, Serializable, transient, serialVersionUID

Module 10: Java 8 Features

Lambda, Functional Interfaces, Predicate, Function, Consumer, Supplier, Method References, Stream API, filter(), map(), flatMap(), reduce(), Optional, Collectors, groupingBy(), partitioningBy(), Parallel Streams, Date & Time API

Module 11: Reflection & Annotations (Advanced Core Java)

Reflection API, Class class, Reflection Methods, Custom Annotations, Built-in Annotations, Marker Interfaces, Cloneable, RandomAccess

Module 12: JVM & Advanced Core Java

Garbage Collection, Heap Generations, Strong/Soft/Weak/Phantom References, JVM Class Loading, Native Methods, JNI (Basic), Dynamic Proxy (Basic), Records, Sealed Classes

Interview Recommendation

Mandatory for freshers: Modules 1–10. Learn Modules 11–12 after gaining confidence or when targeting advanced interviews.